

NISHAL STANISLAUS SILVA, *PhD*

ML Research Engineer | Real-Time & Edge Inference | Audio, Sensor & Time-Series

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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PROFESSIONAL SUMMARY

ML Research Engineer specializing in real-time and on-device inference for audio, sensor and time-series data, with 10+ years shipping constrained embedded systems in modern C++ and Python. Builds the engineering layer around trained models rather than training them: C++17 inference libraries, model-analysis and benchmarking tooling, and latency/CPU profiling tuned to hard compute and power budgets on ARM and embedded Linux hardware. PhD (University of Trento, 2025) on embedded real-time pattern detection, plus 5.5 years at MAS Holdings building C++/Python inference engines and PLC-integrated automation on embedded and industrial hardware. Author of Nebula, an open-source C++17 audio-feature library with per-feature ARM latency benchmarks, and publisher of a detector running at 14 ms on a Raspberry Pi 4 (F1 0.76, 74x faster than a DTW baseline; JAES 2026). Works daily across TensorFlow, PyTorch and ONNX as the consumer-side engineer who has to make a trained model run fast on constrained silicon, with Git, Docker and cross-platform embedded toolchains as the rest of the kit.

CORE COMPETENCIES

ML Engineering: TensorFlow, Keras, PyTorch, ONNX-based model deployment, feature engineering, DSP/MFCC pipelines, model-analysis and benchmarking tooling, ablation-driven accuracy/latency/CPU trade-off design.

Edge & System Optimization: C++17 real-time inference engines, ARM and embedded-Linux CPU profiling, hard latency/compute/power-budget engineering, cross-platform embedded toolchains, PLC and hardware-in-the-loop integration, design-pattern-driven library architecture.

Research Engineering: Benchmark and dataset design, ablation studies, technical writing and peer review, mentoring and code-review practice, cross-domain systems translation.

TECHNICAL STACK

Languages: Python, C++17, C, MATLAB, JavaScript/TypeScript, Shell.

ML / AI: TensorFlow, Keras, PyTorch, scikit-learn, ONNX.

Edge & Systems: Raspberry Pi, ARM CPU profiling, embedded Linux, Elk Audio OS, JUCE, OpenCV, PLC integration.

Data & Cloud: AWS (EC2, S3, ECS, MWAA, RDS), SQL, Pandas/NumPy, ETL pipelines.

Dev & MLOps: Git, Docker, CI/CD, Jenkins, pytest.

PORTFOLIO AND DEMOS (full portfolio: nishal.xyz/research)

- **Nebula:** open-source C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks: github.com/ns2max/nebula
- Real-time embedded pattern detector on Raspberry Pi 4 (MIDI Innovation Awards 2023 finalist): youtu.be/gkOqfOT8zXM

EDUCATION

PhD, Information Engineering and Computer Science – University of Trento, Italy	2021 - 2025
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK	2016 - 2018
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK	2013 - 2014

RESEARCH AND WORK EXPERIENCE

Postdoctoral Researcher | *University of Trento, Trento, Italy*

Jan 2025 - Dec 2025

- Owned the full ML pipeline for streaming audio and IMU/gesture data: acquisition, DSP/feature engineering, training and evaluation, and C++/embedded deployment on Raspberry Pi 4, with monitoring.
- Shipped a real-time detector at 14 ms inference on Raspberry Pi 4, F1 0.76, 31.4% CPU utilization – 74x faster than a DTW baseline (JAES 2026).
- Designed frozen-protocol benchmark suites and ablation studies quantifying accuracy/latency/CPU trade-offs, the same discipline needed to characterize a C++ library across processor targets.

Visiting Researcher | *McGill University, IDMI/CIRMMT, Montreal, Canada*

May 2024 - Aug 2024

- Fused IMU and audio sensor streams via a hierarchical finite-state machine on embedded hardware (Raspberry Pi 4, BNO055 IMU), achieving F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka*

Jan 2024 - Mar 2024

- Conducted a human-centred evaluation of embedded pattern-detection systems (IEEE I3DA 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka*

Aug 2020 - Feb 2021

- Built SQL and Python backend scoring services for financial-forensics anomaly detection running in production.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka*

Jan 2015 - Jul 2020

- Built C++/Python real-time inference engines running on embedded and IoT hardware across manufacturing sites in South Asia, North America and Africa, lifting operational efficiency up to 300%.
- Reverse-engineered PLC trigger points and built three closed-loop automation subsystems in custom C++ for the Aeoon Kyo DTG printer integration.
- Designed and shipped an automated multilingual care-label QC system (OpenCV template and feature matching, conveyor with PLC-controlled reject), cutting inspection time 99.5%.
- Introduced CI/CD and code-review practice; mentored engineers and delivered group-wide computer-vision and embedded-systems training.

SELECTED PUBLICATIONS (full list: nishal.xyz/publications)

- Silva, N. and Turchet, L. *Real-Time Audio Pattern Detection for Smart Musical Instruments*. Journal of the Audio Engineering Society, Vol. 74, No. 3, Mar. 2026. (F1 0.76 at 14 ms on Raspberry Pi 4, 74x faster than baseline)
- Silva, N. and Turchet, L. *Music Information Retrieval as a Service: Latency Characterization of an Edge-to-Server Architecture for Near-Real-Time Audio Analysis*. IEEE International Symposium on the Internet of Sounds, 2026 (accepted).
- Silva, N. and Turchet, L. *A Structural Similarity Index Based Method to Detect Symbolic Monophonic Patterns in Real-Time*. 25th International Conference on Digital Audio Effects (DAFx), Sep. 2022.

INDEPENDENT RESEARCH, HIGHLIGHTS, & RECOGNITION

- **Independent research (Jan 2026 – present):** maintaining Nebula (C++17 audio-feature library, ARM latency benchmarks); MIRaaS paper accepted at IEEE IS2 2026; Smart Drums paper in press at JAES.
- **MIDI Innovation Awards:** Finalist (Sep 2023), Prototypes & Non-Commercial Software category – “Hot Licks” real-time embedded pattern detector.
- **GMOA Recognition (Sri Lanka):** Computer-vision device for COVID-19 patient vitals monitoring deployed across hospital wards.

- Builds real-time ML prototypes for audio and sensor streams under constrained embedded compute, using AI-assisted engineering workflows including Claude Code for tooling and pipeline development.